CSA Recruitment Process: Regression Exercise

The idea is that you solve this entire technical test using the Python programming language. Additionally, you must deliver the file with the script (`.py`) or Jupyter notebook (`.ipynb`) along with all necessary files to run these programs in a compressed folder (`.zip`) to the following emails: `ezequiel.prada@csahavasmn.com`, `tomas.notenson@csahavasmn.com` and `matias.kambourian@csahavasmn.com`. Keep in mind that the file must run without requiring any modifications. The better documented and clearer the code, the better. Try to use the practices you consider most efficient.

On the other hand, you will submit a PDF file solving each of the sections of the two exercises that are not resolved in the other mentioned files (`.py`, `.ipynb` or `.xlsx`). All questions must be answered in English.

Tip: Read the complete test carefully. If you cannot solve a particular section, continue with the next one and then return to think about it once you have solved the others. Try to complete as much as you can.

Part I: Regression

(1) In this exercise we will work with a marketing dataset containing weekly information over 200 weeks of observation. The data is stored in the `'Raw Data'` sheet of the `'Regression-Exercise.xlsx'` file and contains the following time series:

- **Spend Channel 1**: Weekly investment in the first marketing channel
- **Spend Channel 2**: Weekly investment in the second marketing channel
- **Spend Competitors**: Estimated weekly investment by competitors
- **Conversions**: Number of conversions obtained each week

The main objective is to develop a model that allows explaining and predicting conversions (Conversions) using the three marketing investment series as explanatory variables (Spend Channel 1, Spend Channel 2, and Spend Competitors). During the exercise development, you may add additional variables that you consider necessary to improve the model fit.

The simplest model that allows achieving your objective has the following form:

$$Y(t) = \beta_0 + \beta_1 \cdot X_1(t) + \beta_2 \cdot X_2(t) + \beta_3 \cdot X_3(t) + ... + \epsilon(t)$$

where $Y(t)$ is the target variable, $X_i(t)$ is the i-th regressor variable and β_i its associated coefficient.

(a) **Data Exploration and Preparation**:

(i) Load the data from the file and examine the first observations of each time series.

(ii) Perform a basic descriptive analysis (mean, standard deviation, minimum, maximum) of all variables.

(iii) Generate time series plots to identify patterns, trends, or atypical behaviors.

(b) **Modeling**:

(i) Estimate the basic multiple linear regression model proposed in the introduction.

(ii) Plot in the same figure the time series of predicted and actual conversions.

(iii) Do you consider it necessary to add additional variables to the model? Justify your answer and, if so, incorporate the variables you consider appropriate.

(iv) Taking into account that each term of the proposed model[1] has the same units as the target variable, interpret the obtained coefficients and these terms from a business perspective.

(v) Calculate the total incremental conversions associated with each media channel.

(vi) Calculate the yearly incremental conversions associated with each media channel.

(c) **Model Fit Evaluation**:

(i) What metrics would you use to evaluate the goodness of fit of your model? Calculate and interpret these metrics.

(ii) Compare the value of these metrics between the different model versions you have tested.

(d) **Statistical Significance**:

(i) How can you determine if a variable is statistically significant in the regression? Apply this analysis to your final model.

(ii) What conclusions can you draw about the relevance of each marketing channel?

(e) **Model Diagnostics**:

(i) How can you evaluate if there is autocorrelation in your model's residuals? Perform this analysis.

(ii) How can you detect if there is multicollinearity between the regressor variables? Apply the corresponding method.

[1] These terms are called volume

(iii) What implications do these diagnostics have for the validity of your model?

(f) **Econometric Validation**:

(i) Taking into account that each term $\beta_i \cdot X_i(t)$ is called the **volume** of the i-th regressor variable and that the **contribution** is **the percentage of volume represented by each channel**. Calculate the contribution of each variable.

(ii) Assuming that the price associated with each conversion is \$152. Calculate the total revenue associated with each variable.

(iii) Calculate the total annual revenue associated with each variable.

(iv) Taking into account the following definition:

$$\text{CPA} = \frac{\text{Spend}}{\text{Conversions}}$$

Calculate the total and annual CPA for each media variable.

(v) Compare the CPA values for each media channel for the last 2 years. What conclusions can you draw from this comparison?

Part II: Response Curve Optimization

(2) In the `'Response Curve'` sheet of the same file you will find a chart showing the relationship between incremental sales generated by a specific marketing channel and the investment (spend) made in that channel.

This relationship, known as a response curve, is fundamental for marketing budget optimization, as it allows identifying the optimal investment point that maximizes channel profitability. Taking into account that profitability is defined as:

$$\text{ROAS} = \frac{\text{Revenue}}{\text{Spend}}$$

(a) Find that optimal profitability point.

(b) What happens if I increase the spend starting from that optimal spend? And if I decrease it? Interpret.